



Need to model *all* randomness with sigma points

- We now apply sigma-point approach of propagating statistics through a nonlinear function to the state-estimation problem.
- These sigma-points must jointly model *all* randomness:
 - Uncertainty of the state.
 - Uncertainty of process noise.
 - Uncertainty of sensor noise.
- So we first define an augmented random vector x_k^a that combines these random factors at time index k .



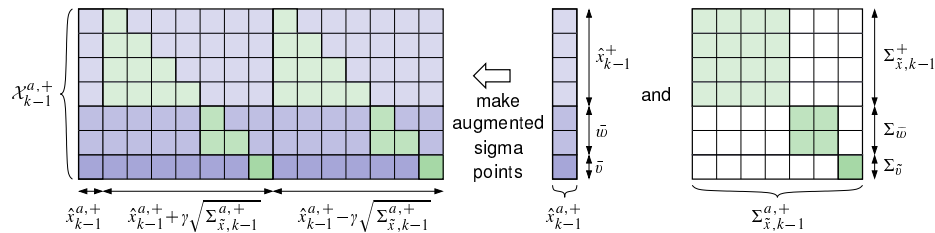
Step 1a: State prediction time update (1)

- First, form augmented prior state prediction, covariance:

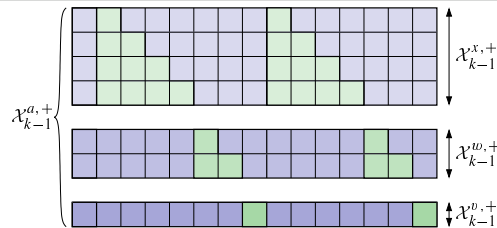
$$\hat{x}_{k-1}^{a,+} = [(\hat{x}_{k-1}^+)^T, \bar{w}^T, \bar{v}^T]^T \text{ and } \Sigma_{\tilde{x},k-1}^{a,+} = \text{diag}(\Sigma_{\tilde{x},k-1}^+, \Sigma_{\bar{w}}, \Sigma_{\bar{v}}).$$
- These factors are used to generate the $p + 1$ augmented sigma points:

$$\mathcal{X}_{k-1}^{a,+} = \left\{ \hat{x}_{k-1}^{a,+}, \hat{x}_{k-1}^{a,+} + \gamma \sqrt{\Sigma_{\tilde{x},k-1}^{a,+}}, \hat{x}_{k-1}^{a,+} - \gamma \sqrt{\Sigma_{\tilde{x},k-1}^{a,+}} \right\}.$$

- Can be organized in convenient matrix form:



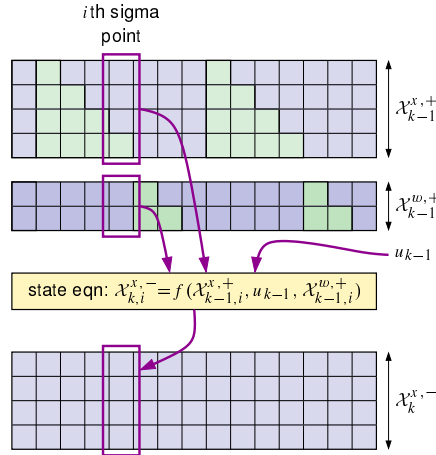
Step 1a: State prediction time update (2)



- Split augmented sigma points $\mathcal{X}_{k-1}^{a,+}$ into state portion $\mathcal{X}_{k-1}^{x,+}$, process-noise portion $\mathcal{X}_{k-1}^{w,+}$, and sensor-noise portion $\mathcal{X}_{k-1}^{v,+}$.



Step 1a: State prediction time update (3)



- Evaluate state equation using all pairs of $\chi_{k-1,i}^{x,+}$ and $\chi_{k-1,i}^{w,+}$ (where subscript i denotes that the i th vector is being extracted from the original set), yielding the prediction sigma points $\chi_{k,i}^{x,-}$.
- That is, compute $\chi_{k,i}^{x,-} = f(\chi_{k-1,i}^{x,+}, u_{k-1}, \chi_{k-1,i}^{w,+})$.



Step 1a-b: State prediction time update (4)

- Finally, present state prediction is computed as:

$$\begin{aligned} \hat{x}_k^- &= \mathbb{E}[f(x_{k-1}, u_{k-1}, w_{k-1}) | \mathbb{Z}_{k-1}] \\ &\approx \sum_{i=0}^p \alpha_i^{(m)} f(\chi_{k-1,i}^{x,+}, u_{k-1}, \chi_{k-1,i}^{w,+}) \\ &= \sum_{i=0}^p \alpha_i^{(m)} \chi_{k,i}^{x,-} = [\chi_k^{x,-}] [\alpha^{(m)}] \end{aligned}$$

■ Can compute with simple matrix multiply.

- Then, covariance estimate is computed as:

$$\begin{aligned} \Sigma_{\tilde{x},k}^- &= \sum_{i=0}^p \alpha_i^{(c)} (\chi_{k,i}^{x,-} - \hat{x}_k^-) (\chi_{k,i}^{x,-} - \hat{x}_k^-)^T \\ &= [\chi_k^{x,-} - \hat{x}_k^-] [\text{diag}(\alpha^{(c)})] [\chi_k^{x,-} - \hat{x}_k^-]^T \end{aligned}$$



Step 1c: Predict system output z_k

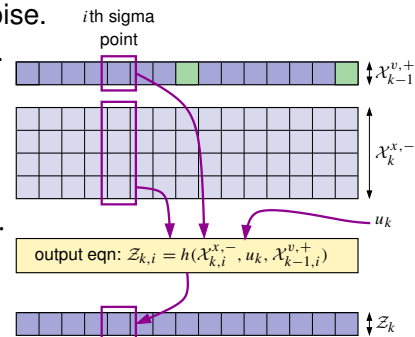
- Output z_k predicted by evaluating model output equation using sigma points describing state and sensor noise.

- First, compute points: $\mathcal{Z}_{k,i} = h(\chi_{k,i}^{x,-}, u_k, \chi_{k-1,i}^{v,+})$.

- Output estimate is then:

$$\begin{aligned} \hat{z}_k &= \mathbb{E}[h(x_k, u_k, v_k) | \mathbb{Z}_{k-1}] \\ &\approx \sum_{i=0}^p \alpha_i^{(m)} h(\chi_{k,i}^{x,-}, u_k, \chi_{k-1,i}^{v,+}) = \sum_{i=0}^p \alpha_i^{(m)} \mathcal{Z}_{k,i}. \end{aligned}$$

- Can be computed with simple matrix multiplication, as was done when calculating $\hat{x}_k^-$ at end of Step 1a.





Step 2a: Estimator gain matrix L_k

- To find L_k , must first compute required covariance matrices:

$$\Sigma_{\tilde{z},k} = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{Z}_{k,i} - \hat{z}_k) (\mathcal{Z}_{k,i} - \hat{z}_k)^T$$

$$\Sigma_{\tilde{x}\tilde{z},k} = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_{k,i}^{x,-} - \hat{x}_k^-) (\mathcal{Z}_{k,i} - \hat{z}_k)^T.$$

- These depend on sigma-point matrices $\mathcal{X}_k^{x,-}$ and $\mathcal{Z}_k$, already computed in Steps 1b and 1c, as well as $\hat{x}_k^-$ and $\hat{y}_k$, already computed in Steps 1a and 1c.
- The summations can be performed using matrix multiplies, as we did in Step 1b.
- Then, we simply compute $L_k = \Sigma_{\tilde{x}\tilde{z},k}^{-1} \Sigma_{\tilde{z},k}^{-1}$.



Step 2b–c: State, covariance measurement update

- Then the state estimate is computed as:

$$\hat{x}_k^+ = \hat{x}_k^- + L_k (z_k - \hat{z}_k).$$

- Finally, the estimation-error covariance matrix is calculated directly from the optimal formulation:

$$\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T.$$



Summary

- SPKF uses sigma-point method to propagate uncertainty of input RV to output of model's (possibly) nonlinear state and output equations.
- Applying this procedure to generic-probabilistic-inference solution yields all six SPKF steps.
- Matrices and vectors are convenient way to store all the sigma points and to compute means and covariances from the sigma points.
- SPKF is now derived!